

Myo-reps Part 1 - The evolution and revolution by Borge Fagerli

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This is the google translated version, with certain changes made by me when google couldn't find the right words. Any mistakes in translation or understanding are entirely mine.

One and a half years have passed since I developed Myo-reps and published the article on this method. The method has this time been thoroughly tested and updated to the ultimate version, which I will present in this series.

I have obviously not been lazy since then and I've had a good chance to get it tested and adjusted in every possible way with a good bit over a hundred different people, both in Norway and abroad. I've also been exchanging experiences, theories and ideas with some key people who are researching the mechanisms of muscle growth, and had many good discussions with Dan Moore, the man behind Max Stimulation, and an inspiration to the first version of the Myo-reps.

I can say with certainty that the current version of the Myo-reps are the most effective training method I have ever tried, and the results of those who have followed my program based on Myo-reps have been formidable. I can go to mention our new Norwegian Champion in Athletic Fitness, André Nilsen, who gained 7kg from his last competition for barely 6 months ago, and has doubled his strength in several of his lifts (he was also best man in strength elements in NM) .

Ayna Kristiansen, our own "pump-Snuppa" which has a vigorous and entertaining blog on XXX realized that she trained too much and ate too little. I put her on a Myo-reps program, doubled the calories, cut down on cardio, and the result was that she has risen 3kg at 10 weeks with a measurable and visible decline in the percentage of fat the first time!

I have a growing list of practitioners with years of training behind them, after 8 weeks on a Myo-reps based program have seen increases of 20-30% in some exercises. I do not know who is most happy to hear such feedback - I as a coach, or he / she that has realized that such increases ...

Yes, yes ... I know this sounds almost evangelical, but I encourage you at least to test it on your own before you come by Janteloven(?) and to tell me that I'm just talking nonsense and rubbish.

I will in this first part of the series of articles go into the theory and performance of Myo-reps, in part 2 we look at the amount of exercise and a unique twist integrated in Myo-reps that are essential for optimal progress, and Part 3 on progression through a training phase. For those who don't like to read as much advanced theory and science, you can go directly to Part 4 which is a simple way to train Myo-reps, and gives examples of training and exercises.

Note that the Myo-reps is a METHOD, not a program, so even if I provide suggestions for exercises and training split in Part 4, then Myo-reps are used in most application setup instead of "traditional" strength training.

I have published numerous articles on programming before and can recommend both the series of articles about Exercise Programming for beginners and advanced, and correct prioritization of muscle groups, Part 1-4 [George notes, if I can find these I will also post them].

Although Myo-reps stand out when you want to increase muscle mass and strength, you should train the same way to KEEP muscle mass and strength on a diet and a calorie deficit. For my own part, I first managed to keep the pressure force on a diet after I developed Myo-reps to the current version - it always tends to drop by 10-20% when the fat starts to get low.

I also want to make clear at once that Myo-reps is NOT suitable for beginners. It is an intensive method that requires that you have correct technique on all exercises, and know your own limitations in terms of performance, exercise tolerance and fatigue threshold. We have an excellent novice program that you can follow, and I would

also recommend at least a half of serious training on a standard 5 × 5 program before considering Myo-reps. Of course you also have enough knowledge about nutrition and calorie needs that you know how to eat to maximize the training effect from Myo-reps.

Let's look at what makes Myo-reps so effective.

Had Arnold trained Myo-reps he would have been greater ... and the president of the United States today.
The fundamental premise: Fiber Activation

In addition to Dan Moore and his MaxStimulation, I will give a great deal of credit for the development of Myo-reps method to the Swedish researcher Mathias Wernbom. He is perhaps best known as one of the main characters behind the comprehensive review of all relevant studies ever published on training for strength and hypertrophy, where the variables exercise frequency, load and volume were summarized. (1)

Wernbom is also the one who has seen the so-called occlusion effect (2), based on the Japanese Kaatsu studies where a tryckmansjett [Ed. Occlusion Cuff?] (similar to that used when measuring blood pressure) around the arms or legs have been able to induce dramatic increases in muscle mass on light weights down to 20% of 1 rep max (1RM), and in some cases just by walking on the treadmill! Similar studies have shown 10-20% increases in muscle cross-section even at advanced lifters after only 2-4 weeks of training (3,4).

It's fairly accepted that one must train with heavy weights to achieve increases in strength and muscle mass, so what is the explanation for this phenomenon? Well, recent reports and studies, where, among other things, the researcher Mathias Wernbom have been involved, in the context of a long list of other studies that make up the pieces of the puzzle, suggests that the maximum fiber activation / recruitment is essential for maximum activation of the signals involved in both muscle growth and adaptation in the nervous system for strength increases.

There are three ways to achieve full activation. (There are more strictly, but these are ones that are of most practical value and relevance for Myo-reps method):

1. Lift a light weight explosively. Also known as speed training. As long as you accelerate the weight maximally, one can get reasonably close to 100% activation. Torque, however, takes over pretty quickly, and you will reflexively hold back lest the bar will be flying into the air. Muscles thus experience maximum activation only in a short pulse (period) in a few milliseconds. One way to extend the impulse is to use elastic bands or machines that progressively increase workload throughout the movement and thus may require you to maintain acceleration.
2. Lift a heavy weight at about 5-6RM or heavier, and attempt to lift as explosively as possible. Although the movement is slow, you will as a result of the load achieve maximum fiber activation. Heavier weights lifted primarily by nerve impulses are coordinated, and not by increased fiber activation [Ed. Heavier weights are lifted by greater nerve and muscle coordination, not by higher fibre activation].
3. Lift light to moderate weight near or to exhaustion. Muscle Fiber Activation follows the so-called "size principle" in which the most endurance adapted fibres and weakest activated first, and then activates the larger and stronger muscle fibers as needed. When you reach the point of exhaustion is generated is not sufficient voltage from the activated muscle fibers that the weight can be lifted. Fatigue can be a neural effect- which includes a reflexive inhibition in the central and peripheral nervous systems of nerve impulses to the muscle to avoid overloading. It will also be varying degrees of metabolic fatigue, the accumulation of H⁺ ions and shortage of ATP (primary energy source that must be recycled before further muscle activity may occur).

To make a long story short, all indications pointing in the direction of No. 3 is the primary reason that so-called occlusion / Kaatsu training is so effective.

Occlusion with a tryckmansjett [Ed. Occlusion Cuff?] reduces blood flow to the muscle, and the physiological response to oxygen deficiency arising (hypoxia) leads to achieving full fiber activation much faster. Thus, we can replicate this effect by training with the continuous span of muscle fatigue, even with light weights but still at least 50-60% load (most able 20-25 reps on this charge).

Wernbom has himself conducted EMG measurements on Myo-reps, and even with loads as light as 50-60% of

1RM measures pulses in fiber activation equivalent to that seen for heavy loads 5RM!

In order to maximize fiber activation, we can therefore conclude that you must avoid resting too long on the top or bottom position of an exercise, lower weight control (2-3 seconds) and then lifting explosively as possible (but still under control). Under certain circumstances it may be an advantage with an isometric contraction or "stop" in the bottom position for varying duration (1-5 seconds), but you must therefore not rest or lose tension in the muscle.

Next point is that we must MAINTAIN max activation so that each successive repetition is "efficient" - that puts all the activated muscle fibers for the load and thus leads to a maximum signal response and the training effect.

How are we doing this? Yes, we take a short break before we continue, and by the rapid recycling of ATP can continue with the series of 1-5 reps of the same strain. It is therefore important to find a correct relation between fatigue and full fiber activation, for we know that too much fatigue will limit the heavy weight we can lift and how many total reps we can expose the muscle. There is thus a delicate balance - just enough to fatigue that we maintain close to 100% fiber activation, but not so much that we can not continue.

Therefore we will from now on be more aware of REPS SPEED, or how explosive we can lift the weight. This in itself is helping to provide full fiber activation, but once the reps, lower rates noticeably from one rep to the next we will begin to tip over the edge into too much exhaustion. If we manage to keep this under this point rather than to transcend it, it's as if we can continue to run in perpetuity reps on some exercises. In the midst of a Myo-reps set, you could actually find that you are MORE explosive than in the beginning, an interesting neurological phenomenon.

This must of course be based on a subjective and honest assessment of both reps speed and how subjectively "hard" you must push yourself to complete the set, and embedded in Myo-reps, is therefore added to something I call the fatigue point, which we return to in part 2.

Let us illustrate the performance of the Myo-reps from start to finish:

- * First set ... and we get to 11 reps with the last two repetitions starting to go slower and heavier. You are now close to full activation of the muscle fiber, and we call this ACTIVATION SET.
- * Put down the weight and breathe deeply in and out 10 times (a = exhale a breath), which is about 20-second pause. This is marked with a +
- * Continue ... 3 explosive reps as far as the latter begin to feel a little heavy/slowing down.
- * Replace the bar / weight and the new 20-second pause.
- * Continue ... 3 reps ... as seen from the weight up, new 20 seconds pause, take off the weight and 3 reps, but you notice that it starts to noticeably slower than on the previous series.
- * Replace the weight and STOP HERE.

It may be advantageous to limit themselves to 15 breaths break (30 seconds), but the heaviest sets towards the end of a training phase where you only run 1 or 2 repetitions in Myo-reps series, so you can actually sleep right up against a minute. As we get closer to more strength-oriented training, and I will come back to this in a later article. Buy RTS manual by Mike Tuchscherer in our webshop if you are impatient and want to read more about it now.

Illustrated more clearly see it like this:

11 reps (near exhaustion) + 3 + 3 reps + 3 + 3 + 3 reps.

= 11+ 15 reps

I write this as 10 to 12 reps in a normal program (or 3 x 10-12), where the number BEFORE + sign means that you will manage approximately 10 to 12 reps before you approach the exhaustion of the first set (set activation). Then there is the short breaks and short sets of 3 reps until you have taken 15 extra reps, which is the number AFTER + sign.

Remember that 10 to 12 15 is only an example, figures may vary depending on where in the training phase you are, exercise, priority, and much more.

Of course you can write up the whole series, but the most important factor for muscle growth, progression, and then load a sufficient number of TOTAL reps at full fiber activation so that the burden is "real" long enough on the muscle. The shorter notation gives us then the information we need to see if we have the desired progress from week to week and month to month.

Rest break between exercises

You can have 2-5 minutes rest between two exercises. To work out an exercise that overlap the same muscle group (s) it is normal that you can not do as many reps on the activation set before reaching exhaustion, but you will still not reduce the weight. The goal of full activation is reached sooner and it's certainly no disadvantage. You can take more rest breaks during the first training sessions so that you get an idea of what loads are correct to start with, and so reduce the breaks eventually.

Myo-reps on the base lift? [Ed. Lower body lifts? Squats, deads etc]

Finally I would just point out that one should exercise caution with Myo-reps on the major lower body lifts front squat, squat and deadlift (and very often bent forward rowing) or at least to limit themselves to 3-5 on the Myo-reps. Most will run out of breath before they can complete a long series of short rest breaks, and the optimal technique is often compromised when training near the point of exhaustion, with increasing risk of damage thereto. It may therefore be better to train with the usual set and longer rest breaks on these exercises. Myo-reps and DC / rest-pause?

Those who have read about the rest-pause method including the highly efficient DC program will likely see great similarities with Myo-reps, but there are some critical differences that you've already seen and will soon see. One is that we'd rather CONTROL fatigue than using it as a goal in itself. To increase the total training volume and frequency are important variables for an optimal training effect. In Part 2 you will be presented with a unique way to control the amount of exercise during the training session so you can find the right balance between rest and stimulation for optimal effect!

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Thanks for Borge for that, you can also see more of his articles on this site here (Parts [1](#), [2](#), [3](#), [4](#)). You can see some of my training programs [here](#).

And now you can get some free coaching for YOUR myo-reps journey. Just fill in the form below and we will get started!

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